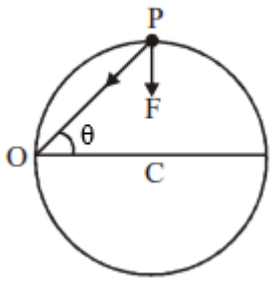
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A particle P is moving on a circle under the action of only one force acting always towards fixed point O on the circumference. Find ratio of $\frac{d^2\theta}{dt^2}$ & $\left(\frac{d\theta}{dt}\right)^2$.



(A) $2\tan\theta$

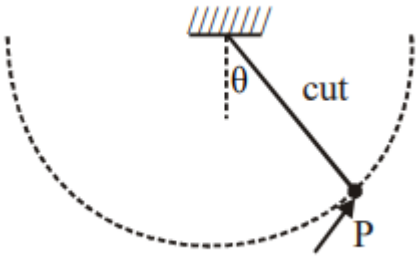
(B) $\tan\theta$

(C) $2\cos\theta$

(D) $2\sec\theta$

Q.2

Consider a pendulum, consisting of a massless string with a mass on the end. The mass is held with the string horizontal, and then released. The mass swings down, and then on its way back up, the string is cut at point P when it makes an angle of θ with the vertical. What should be θ so that the mass travels the largest horizontal distance from P by the time it returns to the height it had when the string was cut?



(A) $\tan^{-1}\sqrt{3}$

(B) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

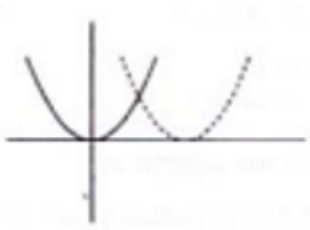
(D) $\tan^{-1}1$

Q.3

A piece of wire is bent in the shape of a parabola $y = kx^2$ with a bead of mass m on it as shown. The bead can

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slide on the wire without friction. It stays at the lowest point of the parabola when the wire is at rest the wire is now accelerated parallel to x-axis with a constant acceleration a. The distance of new equilibrium position of the bead, where the bead can stay at rest with respect to the wire from the y-axis is



(A) $\frac{5a}{4gk}$

(B) $\frac{2a}{3gk}$

(C) $\frac{3a}{4gk}$

(D) $\frac{a}{2gk}$

Q.4

A car is travelling round a section of a race track which is banked at an angle of 15°. The radius of the track is $100\ \mathrm{m}$. What is the speed at which the car can travel without tending to slip ?
$[\tan 15^\circ = 0.2679]$

(A) $12.2\ \mathrm{ms}^{-1}$.

(B) $6.2\ \mathrm{ms}^{-1}$.

(C) $16.2\ \mathrm{ms}^{-1}$.

(D) $8.2\ \mathrm{ms}^{-1}$.

Q.5

A car travelling at $28\ \mathrm{m/s}$ has no tendency to slip on a track of radius $200\ \mathrm{m}$ banked at an angle θ. When the speed is increased to $35\ \mathrm{m/s}$ the car is just on the point of slipping up the track. Calculate the coefficient of friction between the car and the track.

(A) 0.07

(B) 0.18

(C) 0.01

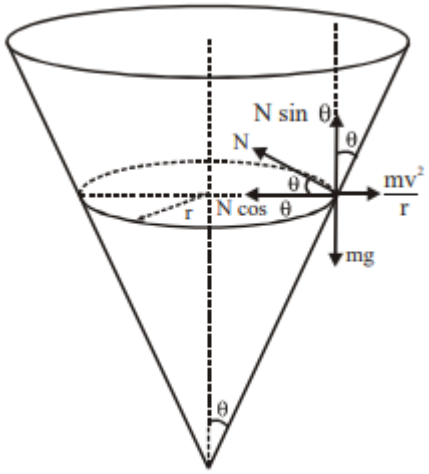
(D) 0.36

Q.6

A particle describes a horizontal circle of radius r in a funnel type vessel of frictionless surface with half one angle θ

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(as shown in figure). If mass of the particle is m, then in dynamical equilibrium find the speed of the particle.



(A) $\sqrt{\frac{rg}{\tan\theta}}$

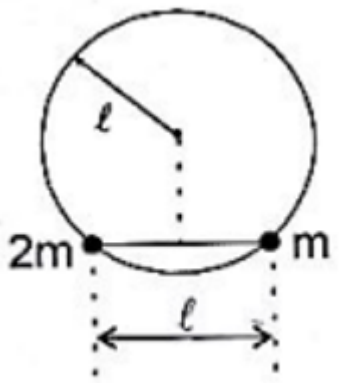
(B) $\sqrt{\frac{rg}{\sec\theta}}$

(C) $\sqrt{\frac{3rg}{\tan\theta}}$

(D) None of these

Q.7

Two beads of mass $2\,m$ and m, connected by a rod of length ℓ and of negligible mass are free to move in a smooth vertical fixed circular wire frame of radius ℓ as shown. Initially the system is held in horizontal position (Refer figure)



Both are given speed v tangentially in anticlockwise direction (when rod is in horizontal position). Find speed v, so that the rod just becomes vertical is :

(A) $\sqrt{\frac{5g\ell}{3}}$

(B) $\sqrt{\left(\frac{3\sqrt{3}-1}{3}\right)g\ell}$

(C) $\sqrt{\frac{3}{2}g\ell}$

(D) None of these

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Q.8

The vertical section of a road over a canal bridge in the direction of its length is in the form of circle of radius 8.9 metre. Find the greatest speed at which the car can cross this bridge without losing contact with the road at its highest point, the center of gravity of the car being at a height $\mathrm{h} = 1.1\ \mathrm{m}$ from the ground. (Take $\mathrm{g} = 10\ \mathrm{m/sec^2}$)

(A) $5\ \mathrm{m/s}$

(B) $7\ \mathrm{m/s}$

(C) $10\ \mathrm{m/s}$

(D) $13\ \mathrm{m/s}$

Q.9

An inextensible string of length $2\ \mathrm{m}$ is fixed at one end A and carries at its other end B a particle of mass $3\ \mathrm{kg}$ which is rotating in a horizontal circle whose centre is $1\ \mathrm{m}$ vertically below A. Choose the correct options -

(A) The angular velocity is $\sqrt{\mathrm{g}}\mathrm{ms}^{-1}$

(B) The tension in the string is $6\ \mathrm{g}\ \mathrm{N}$.

(C) The angular velocity is $2\sqrt{\mathrm{g}}\mathrm{ms}^{-1}$

(D) The tension in the string is $12\ \mathrm{g}\ \mathrm{N}$.

Q.10

A light inextensible string of length 3ℓ is threaded through a smooth ring and carries a particle at each end. One particle A of mass m is at rest at a distance ℓ below the ring. The other particle B of mass M is rotating in a horizontal circle whose centre is A. Choose the correct options -

(A) the angular velocity of B is $\sqrt{g/\ell}$

(B) $\mathrm{m} = \mathrm{M}$

(C) the angular velocity of B is $2\sqrt{\mathrm{g}/\ell}$

(D) $\mathrm{m} = 2\mathrm{M}$

Q.11

A circular race track is banked at 45° and has a radius of $40\ \mathrm{m}$. If the coefficient of friction between the wheels and the track is $1/2$. Choose the correct options -

(A) At speed $20\ \mathrm{m/s}$ a car have no tendency to slip.

(B) the maximum speed at which the car can travel round the track without skidding is $\sqrt{1200}\ \mathrm{m/s}$

(C) At speed $10\ \mathrm{m/s}$ a car have no tendency to slip.

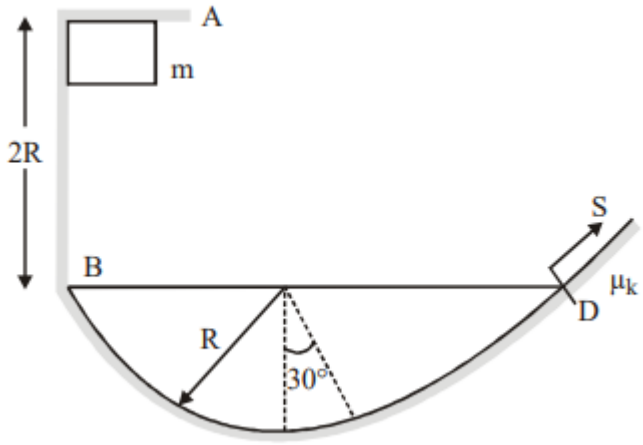
(D) the maximum speed at which the car can travel round the track without skidding is $\sqrt{1100}\ \mathrm{m/s}$

Q.12

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Passage:

A block of mass m is released on a track a shown in figure. The track is smooth from A to D and rough from point D onwards.



Question: The normal force N_B exerted by the track just after crossing point B is $-$

(A) $2mg$

(B) $3mg$

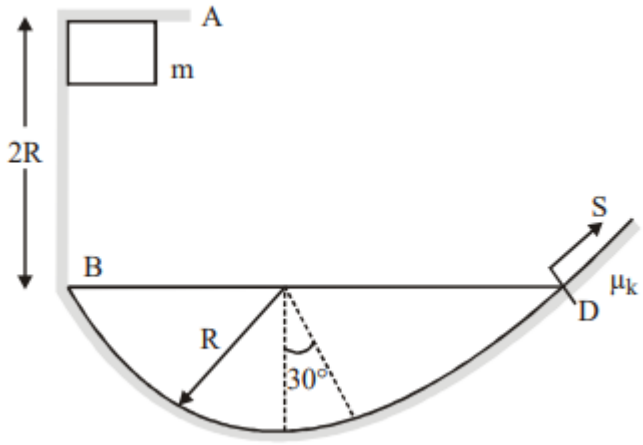
(C) $5mg$

(D) $4mg$

Q.13

Passage:

A block of mass m is released on a track a shown in figure. The track is smooth from A to D and rough from point D onwards.



Question: The normal force N_C at C is $-$

(A) $3mg$

(B) $7mg$

(C) $4mg$

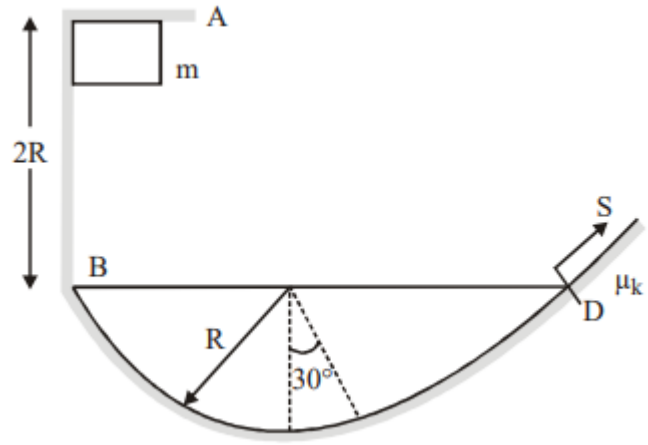
(D) $2mg$

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Q.14

Passage:

A block of mass m is released on a track a shown in figure. The track is smooth from A to D and rough from point D onwards.



Question: The distance travelled along the incline beyond point D where the block stops is $-$

(A) $\frac{4R}{\sqrt{3}\mu_k+1}$

(B) $\frac{2R}{\sqrt{3}\mu_k+1}$

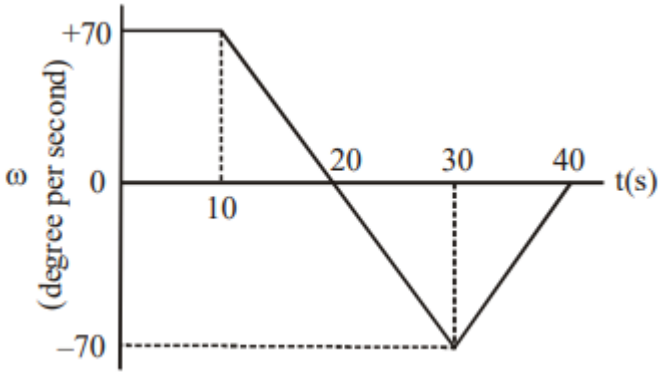
(C) $\frac{4R}{\sqrt{3}\mu_k+3}$

(D) $\frac{4R}{\mu_k+1}$

Q.15

Passage:

Angular speed, ω is the rate at which an object rotates. Although usually measured in radians per second, ω can also be expressed in degrees per second. For instance, consider a merry-go-round. Figure shows its angular speed as a function of time, t. In this graph, a positive value of ω indicates counterclockwise rotation.



Question: A "complete revolution" occurs when the merry-go-round turns through one full circle. Between $t = 0$ and $t = 10$ s, about how many revolutions (approx.) does the merry-go-round complete?

(A) 1

(B) 2

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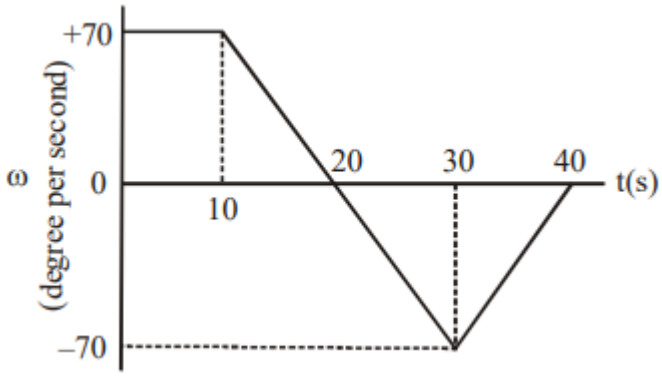
(C) 3

(D) 4

Q.16

Passage:

Angular speed, ω is the rate at which an object rotates. Although usually measured in radians per second, ω can also be expressed in degrees per second. For instance, consider a merry-go-round. Figure shows its angular speed as a function of time, t. In this graph, a positive value of ω indicates counterclockwise rotation.



Question: Between $t = 10$ s and $t = 20$ s Merry-go-round rotates –

(A) with constant rate in clockwise direction.

(B) In clockwise direction and then slow down.

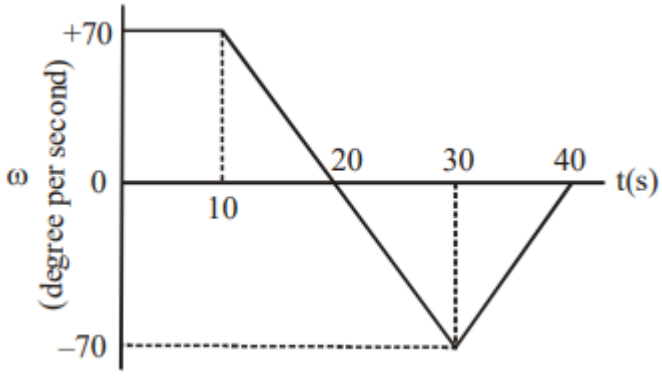
(C) with constant rate in counterclockwise direction.

(D) In counterclockwise direction and then slow down.

Q.17

Passage:

Angular speed, ω is the rate at which an object rotates. Although usually measured in radians per second, ω can also be expressed in degrees per second. For instance, consider a merry-go-round. Figure shows its angular speed as a function of time, t. In this graph, a positive value of ω indicates counterclockwise rotation.



Question: Between $t = 30$ sand $t = 40$ s, the merry-go-round –

(A) rotates clockwise, at a constant rate.

(B) rotates clockwise, and slows down.

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(C) rotates counterclockwise, at a constant rate.

(D) rotates counterclockwise, and slows, down.

Q.18

A small object of mass $0.5\ \text{kg}$ is attached to an end of a massless $2\ \text{m}$ long rope. It is rotated under gravity in a vertical circle with the other end of the rope being at the centre of the circle. The motion is started from the lowest point. Match columns I and II.

Column I	**Column II**
(A) If the speed of the object at lowest point is 3.5m/s.	(p) there will be some point on the circle at which speed of the object is zero but tension in the rope is not zero.
(B) If the speed of the object at lowest point is 8m/s	(q) there will be some point on the circle at which tension in the rope is zero but speed of the object is not zero.
(C) If the maximum tension in the rope is 15N	(r) the object will not be able to reach the highest point.
(D) If the maximum tension in the rope is 30N.	(s) the object will be able to reach the highest point.

(A) A-(p) B-(q) C-(rs) D-(pq)

(B) A-(pr) B-(qr) C-(r) D-(qs)

(C) A-(pq) B-(p) C-(rs) D-(ps)

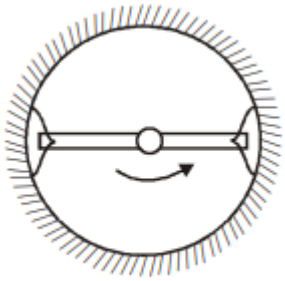
(D) A-(ps) B-(qr) C-(pr) D-(qs)

Q.19

An object is moving in a circle at constant speed v. The magnitude of the rate of change of momentum of the object is proportional to v^n. Find value of n.

Q.20

The essential elements of one form of simple speed governor are as shown to a vertical shaft a horizontal rod is mounted symmetrically and on the horizontal rod are freely sliding brake shoes. When the shaft turns at a frequency of rotation f the brake shoes press against the inner surface of a stationary cylindrical brake drum. If the brake shoes are each of mass m and their thickness dimension is negligible compared to the inner radius of the brake drum and the coefficient of sliding friction between the shoes and the drum is μ, if the power required in watt to turn the governor shaft is $2 \times Xm\mu\pi^3 f^3 r^2$. Find the value of X.



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Answers & Solutions

Q.1 Answer:

—

Solution:

$q_t = F/m \sin\theta$

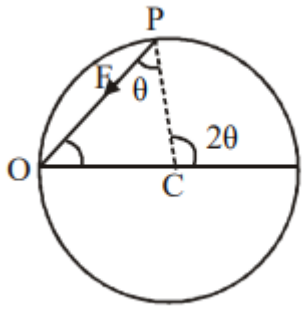
$\frac{Rd^2(2\theta)}{dt^2} = \frac{F}{m}\sin\theta$

$\frac{d^2\theta}{dt^2} = \frac{F\sin\theta}{2mR} \rightarrow (i)$

$a_c = \frac{F}{m}\cos\theta$

$R\left[\frac{d}{dt}(2\theta)\right]^2 = \frac{F}{m}\cos\theta \rightarrow (ii)$

$\frac{\frac{d^2\theta}{dt^2}}{\left(\frac{d\theta}{dt}\right)^2} = 2\tan\theta$



Q.2 Answer:

$\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

Solution:

The speed at angle θ is given by conservation of energy $\frac{1}{2}mv^2 = mgh \Rightarrow \frac{1}{2}mv^2 = mgR\cos\theta$

$\Rightarrow \mathrm{v} = \sqrt{2\mathrm{gR}\cos\theta}$

When string is cut particle moves as a projectile with velocity component. $v_x = v\cos\theta, v_y = v\sin\theta$

The time of flight is $2\left(\frac{v_y}{g}\right)$**. Range is**

$\mathrm{v_x} = \mathrm{v}\cos\theta$

$\mathrm{v_y} = \mathrm{v}\sin\theta$

The time of flight is $\mathrm{t} = 2\left(\frac{\mathrm{v_y}}{\mathrm{g}}\right)$

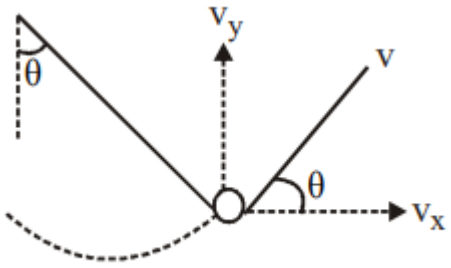
Range is, $x = v_x t = v_x\left(\frac{2v_y}{g}\right) = \frac{2v_x v_y}{g} = \frac{2(v\cos\theta)(v\sin\theta)}{g} = \frac{2\mathrm{v}^2\sin\theta\cos\theta}{\mathrm{g}} = \frac{2(2\,\mathrm{gR}\cos\theta)\sin\theta\cos\theta}{\mathrm{g}} = 4\mathrm{R}\cos^2\theta\sin\theta$

Take the derivative of function of time and equate it to zero.

Thus we get, $0 = 4\mathrm{R}(-2\cos\theta\sin\theta)\sin\theta + \cos^2\theta\cos\theta)$

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or $2\sin^2\theta\cos\theta = \cos^3\theta$ **or** $\tan\theta = \frac{1}{\sqrt{2}}$



Q.3 Answer:

$\frac{a}{2gk}$

Solution:

$$ma\cos\theta = mg\sin\theta$$

$$\frac{a}{g} = \tan\theta = \frac{dy}{dx}$$

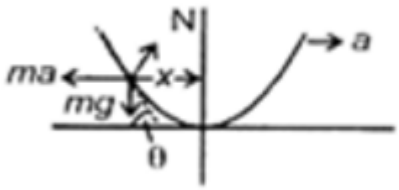
$$\because \quad y = kx^2$$

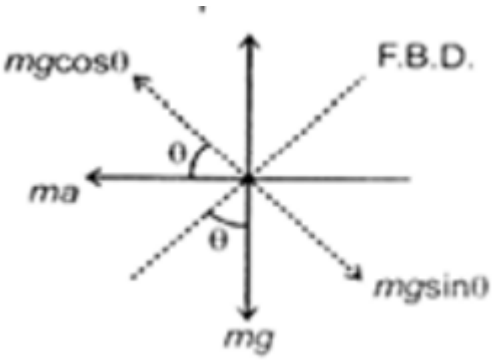
$$\frac{dy}{dx} = 2kx$$

By equation. (i) and (ii)

$$\frac{a}{g} = 2kx$$

$$x = \frac{a}{2kg}$$





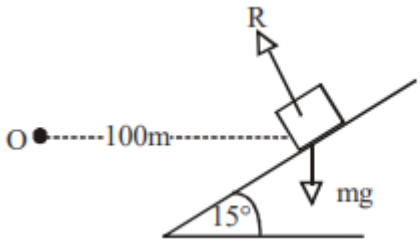
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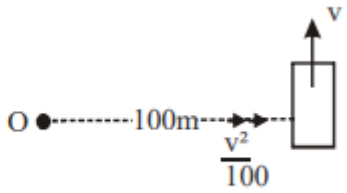
Q.4 Answer:

$16.2\ \mathrm{ms}^{-1}$.

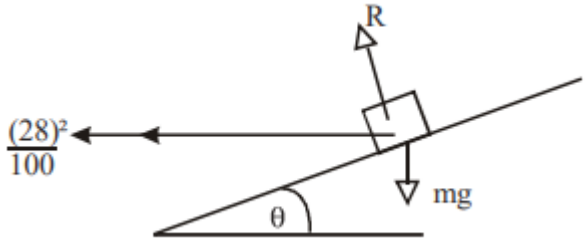
Solution:

If there is no tendency to slip there will be no lateral frictional force.





Horizontally (Newton's law), $R\sin 15^\circ = \frac{mv^2}{100}$



Vertically (no acceleration), $R\cos 15^\circ = mg$

Hence, $\tan 15^\circ = \frac{v^2}{100\,g}$ **i.e.,** $v^2 = 100 \times 9.8 \times 0.2679$

Therefore the design speed is $16.2\ \mathrm{ms}^{-1}$.

Q.5 Answer:

0.18

Solution:

When there is no tendency to slip, no lateral frictional force acts.

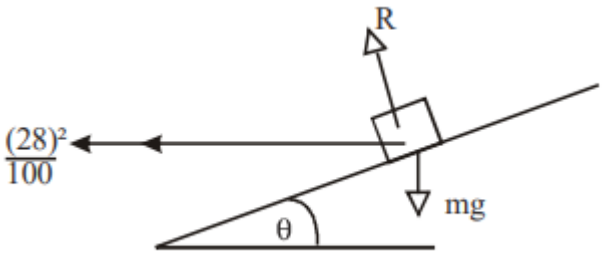
Horizontally $R\sin\theta = m\frac{(28)^2}{200} \ldots (1)$

Vertically $R\cos\theta = mg$ **. . . (2)**

Dividing (1) by (2) gives,

$\tan\theta = \frac{(28)^2}{200\times 9.8} = 0.4$

At a speed higher than $28\ \mathrm{m/s}$ **the car tends to slip outwards and friction acts down the slope.**



Horizontally, $S\sin\theta + \mu S\cos\theta = m\frac{(35)^2}{200} \ldots (3)$

Vertically, $S\cos\theta - \mu S\sin\theta = mg \ldots (4)$

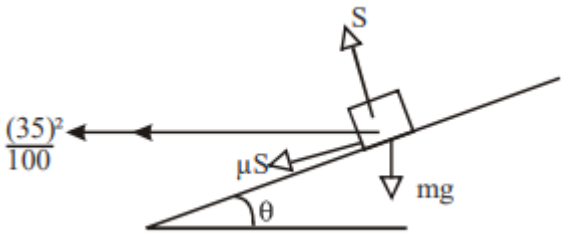
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Dividing (3) by (4) gives $\frac{\sin\theta+\mu\cos\theta}{\cos\theta-\mu\sin\theta}=\frac{(35)^2}{200\times9.8}$

Dividing by $\cos\theta$ **gives** $\frac{\tan\theta+\mu}{1-\mu\tan\theta}=\frac{35\times35\times10}{200\times9.8}$

But $\tan\theta=0.4$ **hence** $\frac{0.4+\mu}{1-0.4\mu}=\frac{5}{8}\Rightarrow3.2+8\mu=5-2\mu\Rightarrow10\mu=1.8$

The coefficient of friction between car and track is 0.18.



Q.6 Answer:

$\sqrt{\frac{\mathrm{rg}}{\tan\theta}}$

Solution:

The normal reaction N **and weight** mg **are the only forces acting on the particle (inertial frame view). The** N **is making an angle** $\left(\frac{\pi}{2}-\theta\right)$ **with the vertical.**

The vertical component of N **balances the weight** mg **and the horizontal component provides the centripetal force required for circular motion.**

$\mathrm{N}\cos\left(\frac{\pi}{2}-\theta\right)=\mathrm{mg}$ **and** $\mathrm{N}\sin\left(\frac{\pi}{2}-\theta\right)=\frac{\mathrm{mv}^2}{\mathrm{r}}$

or $\mathrm{N}\sin\theta=\mathrm{mg}$, **and** $\mathrm{N}\cos\theta=\frac{\mathrm{mv}^2}{\mathrm{r}}$

On dividing we get $\tan\theta=\frac{\mathrm{rg}}{\mathrm{v}^2}$, **so** $\mathrm{v}=\sqrt{\frac{\mathrm{rg}}{\tan\theta}}$

Q.7 Answer:

$\sqrt{\left(\frac{3\sqrt{3}-1}{3}\right)g\ell}$

Solution:

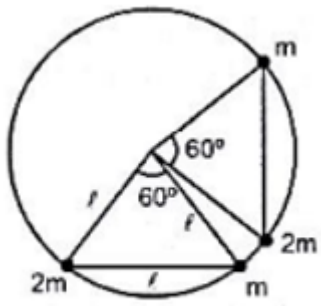
L The speeds given to $2\ \mathrm{m}$ **will also be possessed by** m

$\therefore$ **KE in horizontal position gets converted in PE in vertical position.**

$\frac{1}{2}2mv^2+\frac{1}{2}mv^2=$ **change in** PE **in vertical position.**

$\Delta\mathrm{PE}=2\mathrm{mg}\left[\ell\cos30^\circ-\ell\cos60^\circ\right]+\mathrm{mg}\left[\ell\cos30^\circ+\frac{\ell}{2}\right]$

$2\mathrm{mg}\left[\frac{\ell\sqrt{3}}{2}-\frac{\ell}{2}\right]+\mathrm{mg}\left[\frac{\ell\sqrt{3}}{2}+\frac{\ell}{2}\right]$



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$$\therefore \quad mg\ell[\sqrt{3}-1]-mg\ell\left[\frac{\sqrt{3}+1}{2}\right]$$

$$\Rightarrow mg\ell\left[\sqrt{3}-1+\frac{\sqrt{3}}{2}+\frac{1}{2}\right]=mg\ell\left[\frac{3\sqrt{3}}{2}-\frac{1}{2}\right]$$

$$\text{K.E.}=\frac{1}{2}3\text{mv}^2=mg\ell\left[\frac{3\sqrt{3}-1}{2}\right]$$

$$\therefore v=\sqrt{\left(\frac{3\sqrt{3}-1}{3}\right)g\ell} \quad \text{Ans.}$$

Q.8 Answer:

$10\ \text{m/s}$

Solution:

Let R be the normal reaction exerted by the road on the car. At the highest point, we have $\frac{\text{mv}^2}{(\text{r+h})}=\text{mg}-\text{R}$**, R should not be negative.**

Therefore $v^2 \le (r+a)g=(8.9+1.1)\times 10$

or $\quad \text{v}^2 \le 10\times 10 \Rightarrow \text{v} \le 10\ \text{m/sec}$

$\therefore \quad \text{v}_{\text{max}}=10\ \text{m/sec}$

Q.9 Answer:

The angular velocity is $\sqrt{\text{g}}\text{ms}^{-1}$

Solution:

Vertically (zero acceleration)

$\text{T}\cos\theta=3\ \text{g}...(1)$

Horizontally (Newton's law)

$T\sin\theta=3r\omega^2...(2)$

In triangle $\text{AOB}\begin{cases}\cos\theta=\frac{1}{2}...(3)\\ \text{r}=2\sin\theta...(4)\end{cases}$

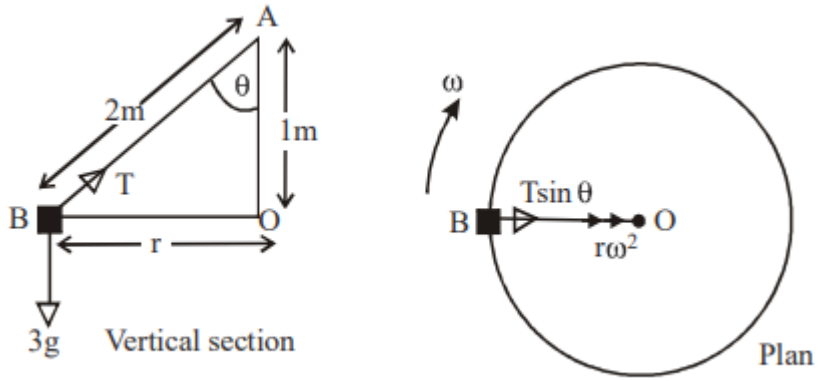
(3) and (4) give $\theta=\frac{1}{3}\pi, \quad r=\sqrt{3}$

[2] \div [1] gives $\tan\theta=\text{r}\omega^2/\text{g}$**. Hence,** $\sqrt{3}=\sqrt{3}\omega^2/\text{g}\Rightarrow\omega^2=\text{g}$

In (1) $\text{T}\times\frac{1}{2}=3\ \text{g}\Rightarrow\text{T}=6\ \text{g}$

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The angular velocity is $\sqrt{g}\,ms^{-1}$ and the tension in the string is $6\ g\ N$.



Q.10 Answer:

the angular velocity of B is $\sqrt{g/\ell}$

Solution:

Since two separate particles are involved we must analyse the state of each. As the ring is smooth, the tension is the same in both sections, BC and AC, of the string.

For mass A (in equilibrium): $T = mg \ldots (1)$

For mass B :

Vertically $T\cos\theta = Mg \ldots (2)$

Horizontally T $\sin\theta = Mr\omega^2 \ldots (3)$

In triangle ABC $\begin{cases} \cos\theta = \ell/2\ell = \dfrac{1}{2} \ldots (4) \\ r = 2\ell\sin\theta \ldots (5) \end{cases}$

(4) and (5) give $\theta = 60^\circ$ **and** $r = \sqrt{3}\ell$

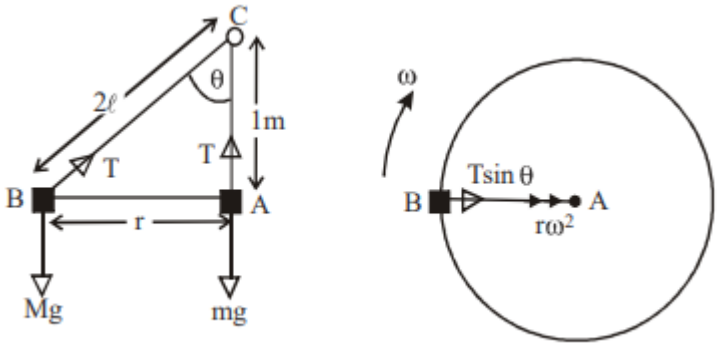
In (2), $T\left(\frac{1}{2}\right) = Mg \Rightarrow T = 2Mg$

But $T = mg$**. Therefore** $m = 2M$

In (3), $T\frac{\sqrt{3}}{2} = M\sqrt{3}\ell\omega^2$

But $T = 2Mg$ **so** $Mg\sqrt{3} = M\sqrt{3}\ell\omega^2$.

Therefore, $\omega = \sqrt{g/\ell}$



Q.11 Answer:

At speed $20\ m/s$ a car have no tendency to slip.

Solution:

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(a) Banking angle is given by $\tan\theta = \frac{v^2}{rg}$

$v^2 = \sqrt{gr\tan\theta} = \sqrt{400} = 20\ m/s$

$f_{smax} = \mu N = \frac{1}{2}\frac{m}{\sqrt{2}}\left(g + \frac{v^2}{r}\right)$

Along the plane, friction $+mg\sin 45^\circ = \frac{mv^2}{r}\cos 45^\circ$

$\frac{m}{2\sqrt{2}}\left(g + \frac{v^2}{r}\right) + \frac{mg}{\sqrt{2}} = \frac{mv^2}{\sqrt{2}r} \Rightarrow \frac{g}{2} + \frac{v^2}{2r} + g = \frac{v^2}{r} \Rightarrow \frac{v^2}{2r} = \frac{3g}{2}$

$\rightarrow -v^2 = 3gr \Rightarrow 3 \times 10 \times 40 = 1200$

Q.12 Answer:

$4mg$

Solution:

Velocity at $B: V_B = \sqrt{2\,g2R} = 2\sqrt{gR}; N_B = m\frac{v_B^2}{R} = \frac{m4gR}{R} = 4mg$

Q.13 Answer:

$7mg$

Solution:

Velocity at $B: V_B = \sqrt{2\,g2R} = 2\sqrt{gR}$; $N_B = m\frac{v_B^2}{R} = \frac{m4gR}{R} = 4mg$

(ii) Velocity at $C: v_C = \sqrt{2\,g3R} = \sqrt{6gR}; N_C = mg + \frac{mv_C^2}{R} = mg + \frac{m6gR}{R} = 7mg$

(iii) $V_D = \sqrt{2\,g2R} = 2\sqrt{gR}, V_f = 0$

$a = -g\left(\sin 30^\circ + \mu_k \cos 30^\circ\right) = g\left(\frac{1}{2} + \frac{\mu_k\sqrt{3}}{2}\right);$

$v_f^2 = v_D^2 + 2as \Rightarrow 0^2 = 4gR$

$2\frac{g}{2}\left(1 + \sqrt{3}\mu_k\right)s \Rightarrow s = \frac{4R}{\left(\sqrt{3}\mu_k + 1\right)}$

Q.14 Answer:

$\frac{4R}{\sqrt{3}\mu_k + 1}$

Solution:

(iii) $V_D = \sqrt{2\,g2R} = 2\sqrt{gR}, V_f = 0$

$a = -g\left(\sin 30^\circ + \mu_k \cos 30^\circ\right) = g\left(\frac{1}{2} + \frac{\mu_k\sqrt{3}}{2}\right); \quad v_f^2 = v_D^2 + 2as \Rightarrow 0^2 = 4gR$

$2\frac{g}{2}\left(1 + \sqrt{3}\mu_k\right)s \Rightarrow s = \frac{4R}{\left(\sqrt{3}\mu_k + 1\right)}$

Q.15 Answer:

2

Solution:

Angle $= 70 \times 10 = 700$

$\Rightarrow \quad \frac{700}{360} = 1.94$, **approx.** $= 2$

Q.16 Answer:

In counterclockwise direction and then slow down.

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Solution:

Counter clockwise slows down.

Q.17 Answer:

rotates clockwise, and slows down.

Solution:

Clockwise slows down.

Q.18 Answer:

A-(pr) B-(qr) C-(r) D-(qs)

Solution:

(A) - pr, (B) - qr, (C) - r, (D) - qs.

$\sqrt{5gr} = \sqrt{5 \times 10 \times 2} = 10\ m/s$

$\sqrt{2gr} = \sqrt{2 \times 10 \times 2} = \sqrt{40}\ m/s$

(A) $3.5\ m/s$ **is less than** $\sqrt{2gr}$**. So the particle will stop before the string becomes horizontal.**

(B) $8\ m/s$ **lies between** $\sqrt{2gr}$ **and** $\sqrt{5gr}$**. The particle will cross the horizontal position but will be able to reach the highest point.**

(C) Tension is maximum at the lowest point. Hence, $15 = mg + \frac{mv^2}{r} \Rightarrow v_2 = \sqrt{40}\ m/s$

In this case particle will just be able to reach the horizontal level. Here both speed and tension become zero.

(D) $30 = mg + \frac{mv_2^2}{r} \Rightarrow v_2 = 10\ m/s$

Here particle will be able to reach the highest point. Tension will become zero at highest point. Hence q **and** s **are matching.**

Q.19 Answer:

—

Solution:

Given : $\frac{dp}{dt} = cv^n$; $\frac{mv^2}{r} = cv^n$; **On comparing,** $n = 2$

Q.20 Answer:

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Solution:

Centripetal force for rotation of brake show comes from normal reaction between shoe and drum.

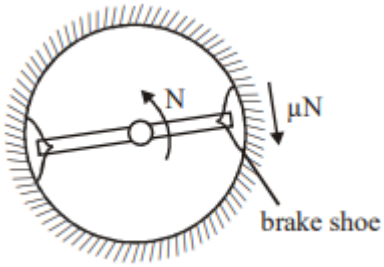
$N = mr\omega^2 = mr(2\pi f)^2$

Friction force $(F) = \mu mr(2\pi f)^2$

Power required to overcome friction force on both the brake shoes,

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$\mathrm{P} = 2\mathrm{Fv} = 2 \times \left[\mu \mathrm{mr}(2\pi \mathrm{f})^2\right] \times \mathrm{r}(2\pi \mathrm{f}) = 16\ \mathrm{m}\mu\pi^3\mathrm{f}^3\mathrm{r}^2 = 2 \times 8m\mu\pi^3 f^3 r^2$. **So** $x = 8$



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